

भारतीय प्रौद्योगिकी संस्थान मण्डी
कमान्द – 175075, जिला – मण्डी,
हिमाचल प्रदेश, भारत



Indian Institute of Technology Mandi
Kamand – 175 075, District – Mandi,
Himachal Pradesh, India

Internship Completion Certificate

This is to certify that **Mr. Nikhil Patyal**, B.tech graduate in Computer Science and Engineering in Artificial Intelligence and Machine Learning student from Jawaharlal Nehru Government Engineering College Sundernagar worked as a research intern on the NMHS-sponsored project titled ***“Integrated Machine Learning and Remote Sensing for Real-Time Predictive Modeling of Rainfall-Induced Landslides in Kamand Valley”*** from 22 June 2026 to 20 July 2026. During this period, Mr. Nikhil Patyal worked under the supervision of **Prof. Varun Dutt** (Principal Investigator and Professor, SCEE, ACS Lab, IIT Mandi) and **Mr. Sahil Sankhyan** (Mentor PhD Scholar, SCEE, IIT Mandi).

As part of the NHMS research project, **Mr. Nikhil Patyal** contributed to the development of an IoT-based landslide monitoring and early warning system integrating environmental sensors, embedded systems, and wireless communication technologies. He was actively involved in circuit interconnection, hardware assembly, ESP32-based sensor integration, sensor testing and calibration, real-time environmental data acquisition, and implementation of the monitoring framework. In addition, he gained practical experience in backend web development using Python, Django, Django REST Framework (DRF), REST API development, Bootstrap-based interfaces, database design, Docker, and Git/GitHub while developing a Data Collection Framework for managing environmental sensor data. The internship strengthened his skills in IoT systems, embedded hardware, backend software development, debugging, collaborative research, and professional engineering practices..

We sincerely appreciate his dedication, technical acumen, and thoughtful engagement with the research objectives.

Date: 20 July 2026

Place: IIT Mandi

Mr. Premchand (Mentor)
M.Tech(R) Scholar
ACS Lab, IIT Mandi

Prof. Varun Dutt, ACS Lab, IIT Mandi

